INHA UNIVERSITY IN TASHKENT

Syllabus

Degree	Bachelor of Science	**Major**	CSE/ICE
Course Title	Engineering Ethics	**Credit**	2
Course ID	NTS4010	**Weekly Class Hours**	2
Instructor	Bobir Muratov		

Course Description

This course is designed to explore the ethical responsibilities of engineers in professional practice. Students will also examine ethical theories, decision-making frameworks, and real-world case studies to navigate ethical dilemmas in engineering. Topics include professional responsibility, safety and risk management, environmental ethics, corporate social responsibility, intellectual property, and the impact of emerging technologies on society. Through discussions, case analyses, and reading of research papers, students will develop the skills to make informed and ethical engineering decisions that align with professional codes of conduct and societal values.

Course Learning Outcomes

By the end of the course, students will be able to:

1. Understand the importance of ethics in engineering practice today.
2. Familiarize with current ethical issues in engineering profession and beyond.
3. Identify, explain, and apply ethical decision making frameworks.
4. Demostrate an understanding of engineer’s role at organizational ethical issues.
5. Develop research, analytical, and communication skills by reading and presenting a paper on current digital and AI ethical issues.

Course Materials

Textbook: Harris, C. E., Pritchard, M. S., James, R. W., Englehardt, E.E., Rabins, M.J. (2019). Engineering ethics concepts and cases (6th edition). Cengage Learning, Inc.

Papers from the books by Oxford University Press on digital and AI ethics.

Grading Policy

Formal assessment will be made on the basis of:

Attendance – required, max. 10 min. late.

Participation – join the discussions during the class.

Paper presentation:

With a team of 5 people, students shall present a paper from the 3 books suggested by the professor.

Books can be found in e-class in Course Summary.

Presentation content can be made based on the sections of papers and beyond.

Each member of the team should submit the PPT file.

For oral presentation, recommended time limit is 20 (min.) – 25 (max.) minutes.

Grading criteria:

Participation of all members (for each missing member, 5 points will be deducted for presenters)

Presentation skills (clear talking and gestures, engagement with audience, content grammar, visuals, logical order, creativity)

Depth of content (appropriate description of information, going beyond the paper)

Quality of discussion (application, lessons learnt, recommendation if any, criticality)

Other (following time limit, presenting the right paper, answering questions of audience)

Deadline: 2nd week

Midterm exam – based on textbook lecture topics (weeks 1-7).

Grade Allocation				
Attendance	Participation	Paper presentation	Midterm	Total
20%	20%	30%	30%	100%

Weekly schedule		
Week	**Content**	**Class**
1	**Theme**	Introduction
	Class Details	Syllabus discussion Engineers: Professionals for the Human Good (Ch.1)
	Seminar	
2	**Theme**	A Practical Ethics Toolkit (Ch.2)
	Class Details	Facts and Consepts Moral Theories The Virtue Ethics
	Seminar	Student presentations of papers
3	**Theme**	Responsibility in Engineering (Ch.3)
	Class Details	Engineering Standards Blame Responsibilitiy and Causation Legal Liability
	Seminar	Student presentations of papers
4	**Theme**	Responsibility in Engineering (Ch.3 cont.)
	Class Details	Harms: Legal Liability and Moral Responsibility Responsibility in Design Impediments to Responsibility
	Seminar	Student presentations of papers
5	**Theme**	Engineers in Organizations (Ch.4)
	Class Details	Engineers and Managers Being Morally Responsible in an Organization Proper Engineering and Management Decisions
	Seminar	Student presentations of papers
6	**Theme**	Engineers in Organizations (Ch.4 cont.)
	Class Details	Responsible Dissent Whistleblowing and Loyalty Cases
	Seminar	Student presentations of papers
7	**Theme**	Trust and Reliability (Ch.5)
	Class Details	Honesty Forms of Dishonesty Why Is Dishonesty Wrong?
	Seminar	Student presentations of papers
8	**Theme**	Midterm exam
	Class Details	

	Seminar	
9	**Theme**	The Engineer's Responsibility to Assess and Manage Risk (Ch.6)
	Class Details	The Engineer's Approach to Risk Difficulties in Determining the Causes and Likelihood of Harm The Public s Approach to Risk
	Seminar	Student presentations of papers
10	**Theme**	The Engineer's Responsibility to Assess and Manage Risk (Ch.6 cont.)
	Class Details	Communicating Risk and Public Policy The Engineer s Liability for Risk Becoming a Responsible Engineer Regarding Risk
	Seminar	Student presentations of papers
11	**Theme**	Engineering and the Environment (Ch.7)
	Class Details	Development of the Modern Environmental Movement Environmental Law and Policy Life Cycle Analysis
	Seminar	Student presentations of papers
12	**Theme**	Engineering and the Environment (Ch.7 cont.)
	Class Details	Sustainable Development and Engineering Practice Business and Sustainable Development Cultivating the Virtue of Respect for Nature
	Seminar	Student presentations of papers
13	**Theme**	Engineering in the Global Context (Ch.8)
	Class Details	The Movement Toward Globalized Engineering Educational Standards Ethical Resources for Globalized Engineering Economic Underdevelopment
	Seminar	Student presentations of papers
14	**Theme**	New Horizons in Engineering (Ch.9)
	Class Details	Environmental Responsibility and Sustainable Development Autonomous Vehicle Development Internet of Things, Big Data, and Cyber Security
	Seminar	Student presentations of papers
15	**Theme**	Final exam week – no class
	Class Details	
	Seminar	